

Low Code EXPERIMENT-1

Comparative study of low-code development versus traditional application development approaches.

THAMBURAJ B

240701559

AIM

To compare low-code development with conventional application development, examining their development methods, time requirements, expenses, level of complexity, adaptability, and overall ease of implementation.

INTRODUCTION TO LOW CODE DEVELOPMENT

Low-code development is a modern approach to creating software applications with minimal manual coding. It uses visual programming environments, drag-and-drop elements, reusable components, pre-designed templates, and built-in features to make development faster and simpler.

Low-code platforms allow developers to create interfaces, workflows, databases, and system integrations using graphical tools instead of writing large amounts of code. This reduces development time and enables both professional developers and users with basic programming knowledge to build applications.

Low-code technology is useful for developing business applications, prototypes, automation systems, and simple to

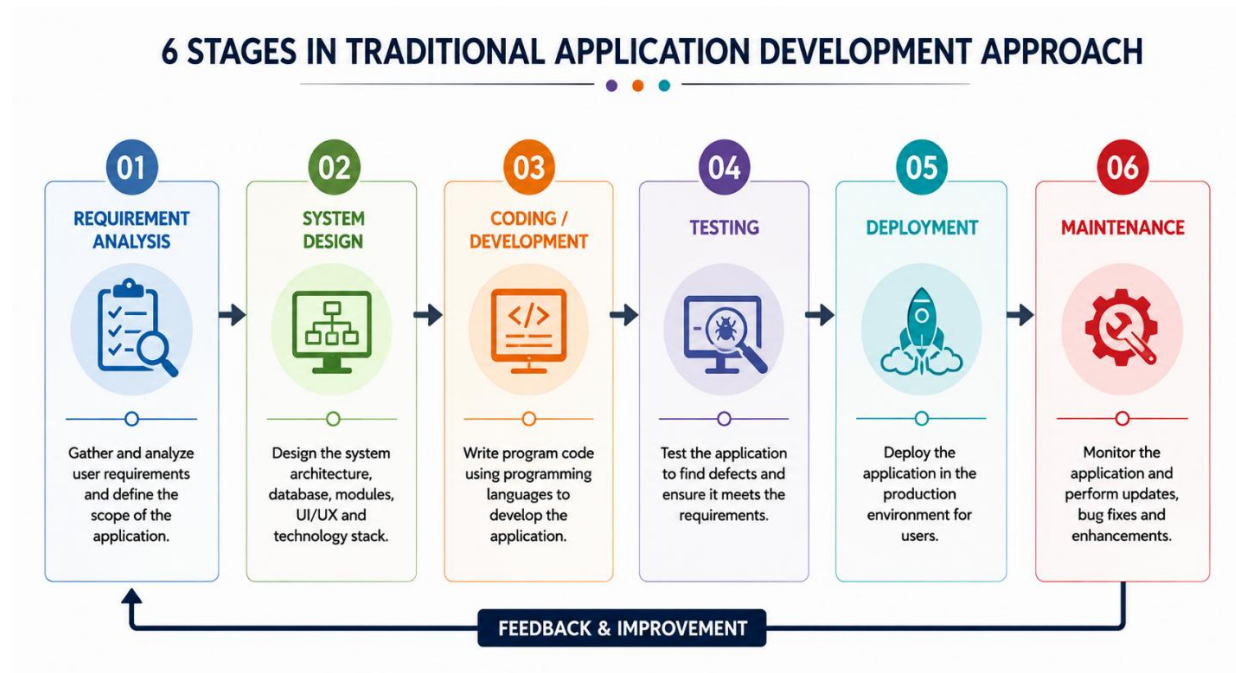
moderately complex solutions. However, it may have limitations in flexibility, customization, and advanced functionality compared with traditional software development.

INTRODUCTION TO TRADITIONAL APPROACH

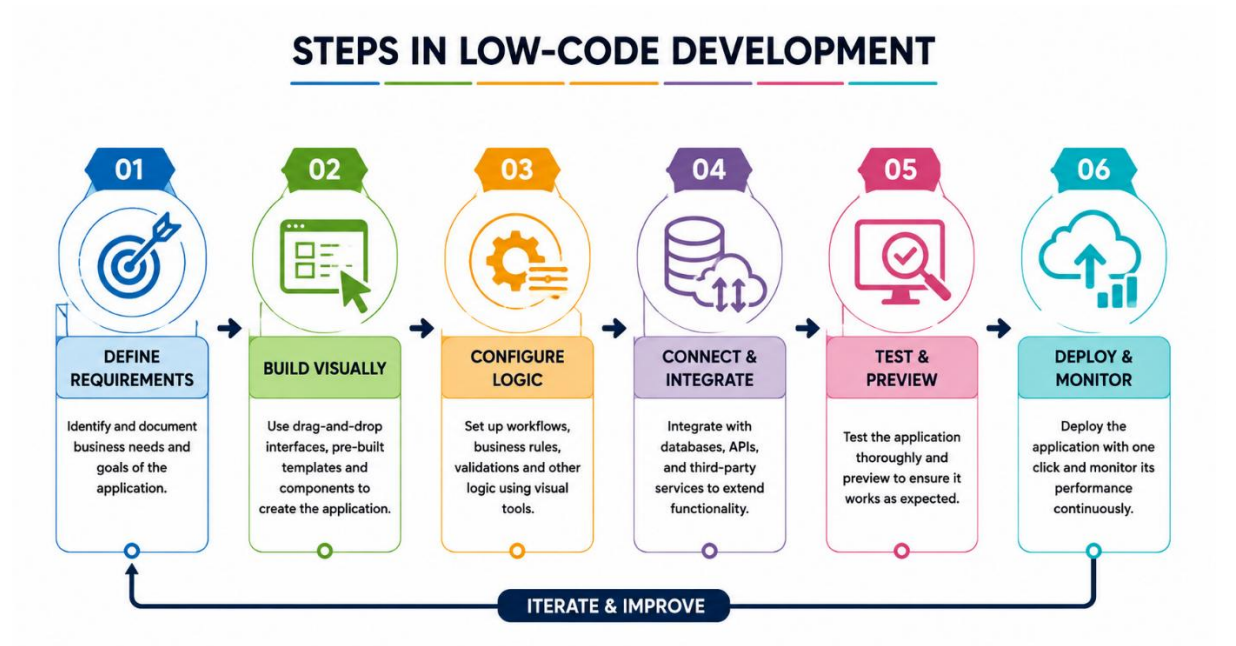
Traditional application development is a standard method of creating software applications by manually writing source code using programming languages such as Java, Python, C++, or C#. It usually follows a systematic software development life cycle, starting with requirement gathering and system planning, followed by coding, testing, implementation, and maintenance.

In this method, developers have full control over the application's features, architecture, interface, database, and external integrations. However, it requires experienced programmers and generally demands more time and resources for application development and maintenance. Traditional development is well suited for complex and highly customized applications where greater flexibility, detailed customization, and complete control over the system are essential.

STAGES IN TRADITIONAL APPROACH



STEPS IN LOW CODE



COMPARITIVE ANALYSIS FOR TRADITIONAL AND LOW CODE

Criteria	Traditional Development	Low-Code Development
Development Method	Involves writing application code manually using programming languages.	Uses visual interfaces, drag-and-drop elements, and minimal manual coding.
Development Time	Generally requires more time for planning, coding, testing, and implementation.	Reduces development time through reusable components and ready-made templates.
Programming Skills	Requires advanced programming skills and technical expertise.	Requires basic programming knowledge with less coding experience.
Customization	Offers greater flexibility and allows extensive customization.	Customization depends on the features and limitations of the platform.
Cost and Maintenance	Usually involves higher development costs and greater maintenance effort.	Helps reduce development expenses and makes maintenance easier.

RESULT

The comparison between traditional and low-code development was successfully analyzed. Traditional development offers greater control, flexibility, and customization, but it generally demands more time, effort, and programming expertise. In contrast, low-code development supports quicker application creation with reduced coding and easier maintenance. Therefore, the appropriate development approach can be chosen according to the project’s complexity, requirements, time constraints, and available resources.

